

Section 3.8 Exponential Growth and Decay

39) Let $A(t)$ be the amount of money in an account where \$1000 was invested and 8% interest is paid compounded continuously. Find and explain each of the following: $A(10)$, $A'(10)$ and $A''(10)$.

$$A = Pe^{rt}$$

$$A = 1000 e^{0.08t}$$

$$A = 1000 e^{0.08(10)}$$

$$A = \$2,225.54$$

$$A' = 1000 e^{0.08t} (0.08)$$

$$A'(10) = 1000 e^{0.08(10)} (0.08)$$

$$A'(10) = \$178.04$$

Account grows by \$178.04 / yr
after 10 years.

$$A'' = 1000 e^{0.08t} (0.08)(0.08)$$

$$A''(10) = 1000 e^{0.08(10)} (0.08)(0.08)$$

$$A''(10) = \$14.24$$

The growth rate is accelerating
by \$14.24 / yr when $t = 10$.

40) A bacteria culture grows with a constant relative growth rate. After 2 hours there are 800 bacterial and after 10 hours there are 80,000 bacteria. Find an expression for the size of the population after t hours.

$$A = Pe^{rt}$$
$$A = 800e^{r(8)}$$

$$\frac{80,000}{800} = \frac{800e^{r(8)}}{800}$$

$$100 = e^{r(8)}$$
$$\ln(100) = \ln(e^{r(8)})$$
$$\ln(100) = r(8)$$

$$r = \frac{\ln(100)}{8}$$

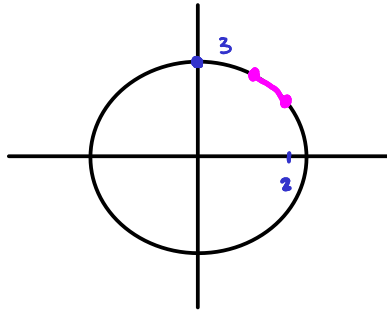
$$r = 0.576$$

$$A = 800e^{0.576t}$$

Section 3.9 Related Rates

41) An object is travelling on the path $x^2 + y^2 = 9$ with $\frac{dx}{dt} = -10 \text{ m/s}$ when $x = 2$.

Find $\frac{dy}{dt}$ and explain when it has to do with the object.



$$2x \frac{dx}{dt} + 2y \frac{dy}{dt} = 0$$

$$2y \frac{dy}{dt} = -2x \frac{dx}{dt}$$

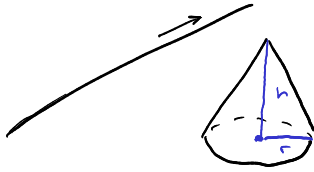
$$\frac{dy}{dt} = \frac{-\cancel{2}x \frac{dx}{dt}}{\cancel{2}y}$$

$$\begin{aligned}x^2 + y^2 &= 9 \\2^2 + y^2 &= 9 \\y^2 &= 5 \\y &= \sqrt{5}\end{aligned}$$

$$\frac{dy}{dt} = \frac{-2(10)}{\sqrt{5}}$$

$$\frac{dy}{dt} = 8.944 \text{ m/s}$$

42) Gravel is being dumped from a conveyor belt at a rate of $40 \text{ ft}^3 / \text{min}$. The coarseness of the gravel is such that the pile formed is a cone whose base diameter and height are always the same. How fast is the height of the pile increasing when the pile is 10 feet tall?



$$V = \frac{1}{3} \pi r^2 h$$

$$2r = h$$

$$r = h/2$$

$$V = \frac{1}{3} \pi (h/2)^2 h$$

$$V = \frac{1}{3} \pi \frac{h^3}{4}$$

$$V = \frac{\pi}{12} \cdot h^3$$

$$\frac{dV}{dt} = 40$$

$$\frac{dV}{dt} = \frac{\pi}{12} \cdot 3h^2 \cdot \frac{dh}{dt}$$

$$h = 10$$

$$40 = \frac{\pi}{4} (10)^2 \cdot \frac{dh}{dt}$$

$$160 = \pi (100) \frac{dh}{dt}$$

$$\frac{dh}{dt} = \frac{160}{100\pi} = \frac{1.6}{\pi} \approx 0.509 \text{ ft/min}$$

Section 3.10 Differentials

43) Compute dy and Δy given that $\Delta x = 1$ and $x = 1$ for the function $y = \sqrt{x}$.

$$\Delta x = 1$$

$$y = \sqrt{1} = 1$$

$$y = \sqrt{2} = 1.414$$

$$\Delta y = 1.414 - 1 = 0.414$$

$$\Delta x \cdot \frac{dy}{dx} = \frac{1}{2\sqrt{x}} \cdot \Delta x$$

$$dy = \frac{1}{2\sqrt{1}} \cdot 1$$

$$dy = 0.5$$

44) The volume of a cube is given to be 80 ft^3 with a maximum error in side measurement to be 0.01 feet. Use differentials to estimate the possible error in the volume.

$$x = \text{length side}$$

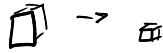
$$V = x^3$$

$$\frac{dV}{dx} = 3x^2$$

$$dV = 3x^2 \cdot dx$$

$$dV = 3(\sqrt[3]{80})^2 \cdot (0.01)$$

$$dV = 0.557 \text{ ft}^3$$



$$80 = x^3$$

$$\sqrt[3]{80} = x$$

Chapter 3 Review

45) Find $\frac{dy}{dx}$ for each of the following functions:

a) $y = \tan(\tan x)$ b) $y = \tan x(\tan x)$ c) $y = e^{(\tan x)}$ d) $y = \tan(e^x)$

a) $y' = \sec^2(\tan x) \cdot \sec^2 x$

b) $y' = \sec^2 \cdot \tan x + \tan x \cdot \sec^2 x$
 $y' = 2 \sec^2 x \cdot \tan x$

c) $y' = e^{\tan x} \cdot \sec^2 x$

d) $y' = \sec^2(e^x) \cdot e^x$

46) Find the equation of the line tangent to $x^2 + 4xy + y^2 = 13$ at the point (2,1).

$$2x + 4y + 4xy' + 2y \cdot y' = 0$$

$$2(2) + 4(1) + 4(2)y' + 2(1)y' = 0$$

$$10y' = -8$$

$$y' = -8/10$$

$$y' = -0.8 \quad \text{Slope}$$

$$y-1 = -0.8(x-2)$$

$$y-1 = -0.8x + 1.6$$

$$y = -0.8x + 2.6$$

47) Find any point where $\frac{dg}{dx} = 0$ for the function $g(x) = \tan^{-1}(\sin^{-1} \sqrt{x})$.

$$g'(x) = \frac{1}{1 + (\sin^{-1} \sqrt{x})^2} \cdot \frac{1}{\sqrt{1 - (\sqrt{x})^2}} \cdot \frac{1}{2\sqrt{x}}$$

$$0 = \frac{1}{(1 + (\sin^{-1} \sqrt{x})^2) \sqrt{1-x} \cdot 2\sqrt{x}}$$

No Solution!

$x=1$ and $x=0$ is where
the derivative is undefined.

Section 4.1 Maximum and Minimum Values

48) Find all minimum and maximum values for $f(x) = x^3 - 5x + 6$.

$$f'(x) = 3x^2 - 5$$

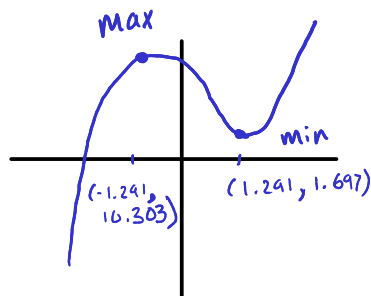
$$0 = 3x^2 - 5$$

$$5 = 3x^2$$

$$\frac{5}{3} = x^2$$

$$\pm \sqrt{\frac{5}{3}} = x$$

$$x = \pm 1.291$$



49) Find all minimum and maximum values for $f(x) = x - \ln x$.

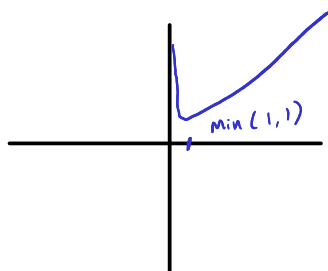
$$f'(x) = 1 - \frac{1}{x}$$

$$f'(x) \text{ DNE at } x=0$$

$$0 = 1 - \frac{1}{x}$$

$$\frac{1}{x} = 1$$

$$1 = x$$



50) Find all minimum and maximum values in the interval $[0, 2\pi)$ for $y = \cos(2x) + \sin x$.

$$y' = \underbrace{-\sin(2x)}_{\downarrow} \cdot 2 + \cos x$$

$$y' = -2 \cdot 2 \sin x \cdot \cos x + \cos x$$

$$0 = \underbrace{\cos x}_{\downarrow} \underbrace{(-4 \sin x + 1)}_{\downarrow}$$

$$\cos x = 0$$

$$\begin{aligned} x &= \pi/2 \\ x &= 3\pi/2 \end{aligned}$$

$$-4 \sin x + 1 = 0$$

$$\sin x = \frac{1}{4}$$

$$x = \sin^{-1}\left(\frac{1}{4}\right)$$

$$x = 0.253$$

$$x = \pi - 0.253$$

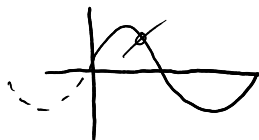
$$x = 2.889$$

$$\text{MAX}^s (0.253, 1.125), (2.889, 1.125)$$

$$\text{MIN}^s (1.571, 0), (4.712, -2)$$

Section 4.2 The Mean Value Theorem

51) Show that the equation $2x - 1 = \sin x$ has exactly one real root.



$$2x - 1 - \sin x = 0$$

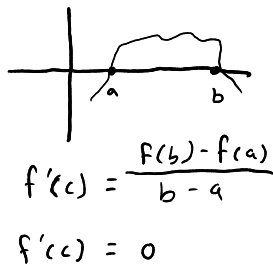
$$f(x) = 2x - 1 - \sin x$$

$$f'(x) = 2 - \cos x$$

$$0 = 2 - \cos x ?$$

$$\cos x = 2 ?$$

Not possible



52) At 2:00PM a car's speedometer read 30 mi/h and at 2:10PM it read 50 mi/h . Prove that at some time between 2:00 and 2:10, the car was accelerating at 120 mi/h^2 .

$$v(0) = 30 \quad \text{and} \quad v(10) = 50$$

$$v'(t) = \frac{50 - 30}{\frac{(10 - 0)}{60}}$$

$$v'(t) = \frac{20}{1/6}$$

$$v'(t) = 20 \cdot \frac{6}{1}$$

$$v'(t) = 120 \text{ mi/h}^2$$

$$\text{Acceleration} = 120 \text{ mi/h}^2$$